

Lab Class: Physics

Quantitative measurement of the specific heat capacity and heat of fusion of a eutectic alloy

Niranjana Parashar, Nahom Gebreyohannis, Gulnara Amangeldiyeva

SS 2025

1 Introduction

The purpose of the experiment was to determine two important thermal properties of a eutectic alloy (Wood's metal): the specific heat capacity and the heat of fusion. These properties help us understand how such materials can be used in practical settings, such as energy storage or thermal regulation. We heated the sample, measured how the temperature changed over time, and used that to analyze the temperature-voltage relationship.

2 Theory

The specific heat capacity c is the property that implies the energy required to raise the temperature of a unit mass of a substance by 1 K and is given by:

$$c = \frac{1}{m} \cdot \frac{dQ}{dT} \quad (1)$$

When a sample containing crucible is heated, it experiences a heat capacity of both the sample and the crucible combined. So, the relation can be expressed as:

$$C_{tot} = C_{sample} + C_{cont} \quad (2)$$

The thermoelectric voltage U_{th} of a thermocouple is approximated by a second-order polynomial:

$$U_{th} = a \cdot (T - T_{ref})^2 + b \cdot (T - T_{ref}) + c \quad (3)$$

For this experiment the reference temperature was set to be the melting point of ice (0°C).

During the melting phase, the temperature remains constant, and the energy supplied is used to break the bonds of the solid. The temperature of the sample increases over time when a constant electrical heating power is applied. This relation is given by:

$$T = T_u + P_{heat} \cdot R_{th} \cdot (1 - e^{\frac{-t}{C_{tot} \cdot R_{th}}}) \quad (4)$$

where R_{th} is the thermal resistance of the system and T_u is the room temperature.

For the heat of fusion, we analyze the energy required to change the phase of the material from solid to liquid without a change in temperature. This shows up as a plateau on the temperature-time graph.

The heating power P_{heat} supplied to the sample is partially lost through the insulation. So, the formula of the heat of fusion is given by:

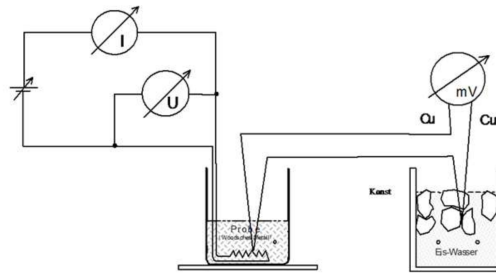
$$Q_m = (t_2 - t_1) \cdot (P_{heat} - P_{loss}) \quad (5)$$

where t_1 and t_2 are the start and end time of the melting phase.
The specific heat of fusion will then be:

$$q_m = \frac{Q_m}{m} \quad (6)$$

3 Experimental Setup

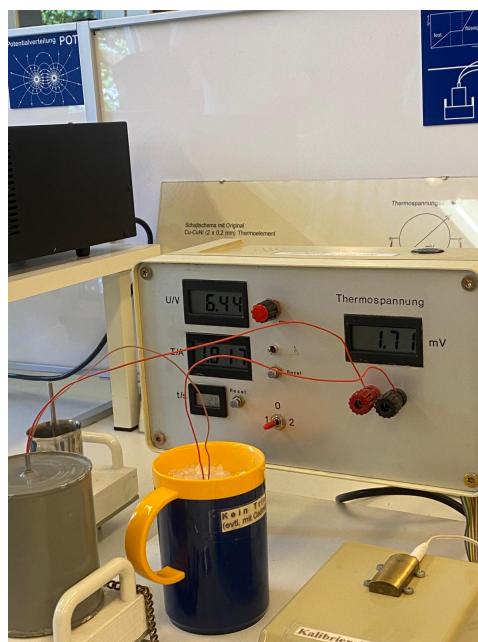
The experiment was set up in a laboratory provided by the *Munich University of Applied Sciences*, which is used for a variety of thermodynamics experiments.



(7)

The setup included:

- Voltmeter and ammeter to monitor heating power
- Power supply
- Crucible with Wood's metal sample
- Thermocouple for temperature measurement
- Built-in stopwatch
- Ice bath



(8)

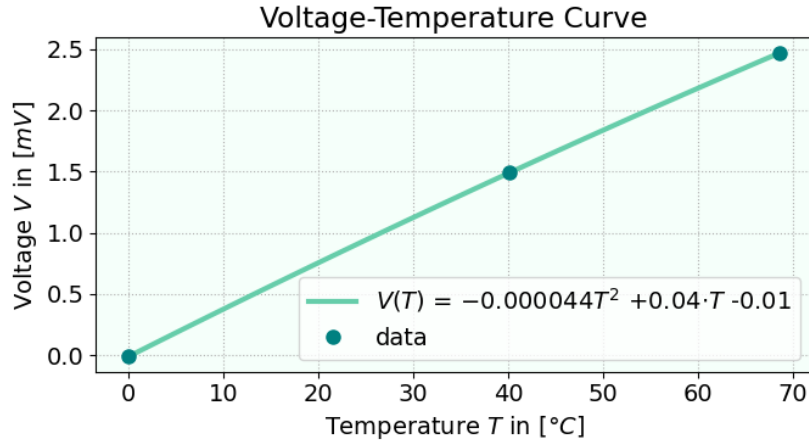
4 Experimental Produce

The measurements were executed using the setup as follows.

1. **Calibration:** First the thermocouple was placed in the ice bath to control that the value ($V_0 = -0.01mV$) read close to $0mV$, verifying it corresponded to $0^\circ C$ (T_0). The heater for the temperature-controlled bore was then switched on and set to $40.1^\circ C$ (T_1). This yielded a thermoelectric voltage of $1.49mV$ (V_1). Then once again the heater temperature was changed and switched $68.6^\circ C$ (T_2). This gave a thermoelectric voltage of $2.47mV$ (V_2).
2. **Heating:** The thermocouple was inserted into the sample and the heater turned on. The current was set to $1.818 \pm 0.019A$ and a heating voltage of $6.44 \pm 0.04V$ was read. Every 40 seconds, the thermoelectric voltage generated by the thermocouple was recorded.
3. **Data Collection:** The voltage readings were recorded throughout the heating process. These values were later converted into temperatures using a quadratic calibration formula derived from earlier reference points.

5 Analysis

To convert voltage readings from the thermocouple into temperature, the initial three values of temperature and corresponding voltage were used to fit a second-order polynomial equation.



(9)

The equation obtained from this was then solved for the temperature T to get the conversion formula needed. Once we found the coefficients we were able to calculate temperatures of every point.

$$T(V) = \frac{-b + \sqrt{b^2 - 4 \cdot a \cdot (c - V)}}{2 \cdot a} \quad (10)$$

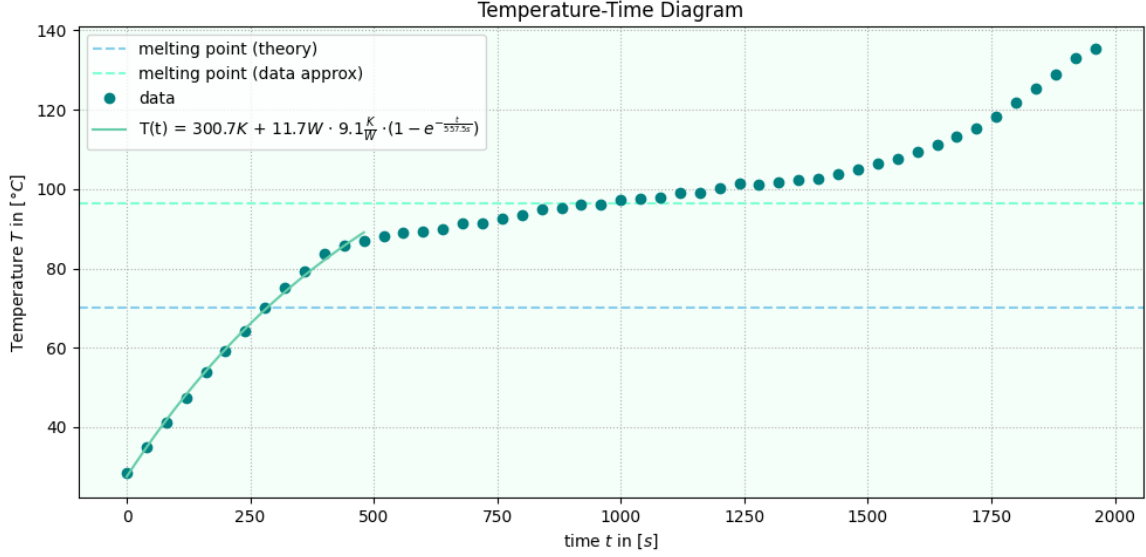
where $a \approx -4.4 \cdot 10^{-5} \frac{mV}{C^2}$, $b \approx 0.04 \frac{mV}{C}$ and $c \approx -0.01mV$ obtained from the curve fit.

The resulting temperature values were analyzed using a temperature-time diagram. By curve fitting equation (4) to this diagram we were able to determine the following quantities:

- $T_u = 300.66 \pm 0.8K$ or $27.51 \pm 0.8^\circ C$ (the ambient temperature)
- $R_{th} = 9.1 \pm 0.8 \frac{K}{W}$ (the thermal resistance)

- $C_{tot} = 61.2 \pm 3.3 \frac{J}{K}$ (total heat capacity)

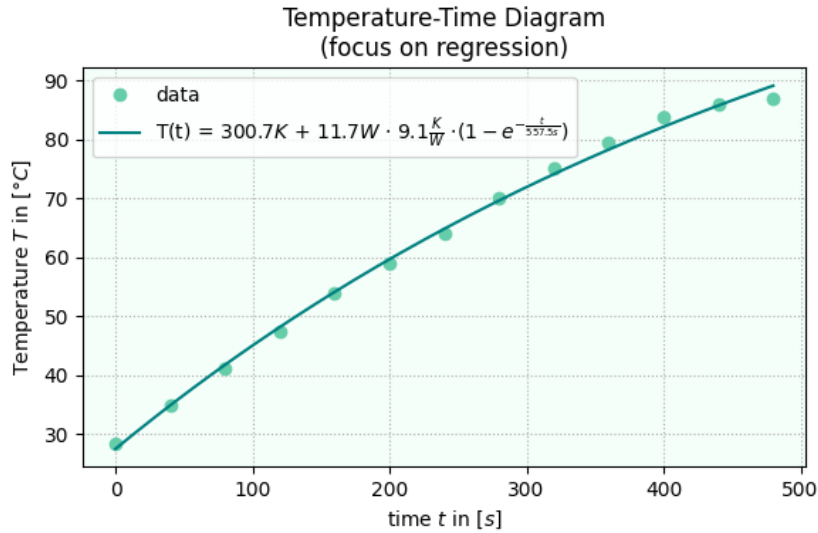
$P_{heat} = 11.7 \pm 0.14W$, from the heating voltage and heating current ($U_{heat} \cdot I_{heat}$).



(11)

The melting point ($95.92 \pm 4.3^\circ C$) was obtained by approximating the points, where the slope changed and averaging the values in between. This gives a value which is off by 37% from theory ($70^\circ C$).

Using the melting point approximated with the measurement data, the equation for the melting process was fitted, using equation



(12)

Once we found the total heat capacity, we were able to calculate the heat capacity of the sample by subtracting the heat capacity of the container.

$$C_{sample} = C_{tot} - C_{cont} \quad (13)$$

where $C_{cont} = 0.48 \frac{J}{g \cdot K} \cdot 48.8g = 23.42 \pm 0.096 \frac{J}{K}$, using given values.

This gives us a value of $C_{sample} = 37.82 \pm 3.3 \frac{J}{K}$. Therefore, the specific heat capacity was calculated by dividing this value by the mass of the sample, $c_{sample} = \frac{C_{sample}}{mass} = \frac{37.82 \frac{J}{K}}{0.33kg} = 114.6 \pm 10 \frac{J}{kg \cdot K}$. This has a deviation of 33.4% from the literature value $172 \frac{J}{kg \cdot K}$. To determine the specific heat of fusion, we used equation (5). We used $t_1 = 520s$ and $t_2 = 1440s$ by analyzing the Temperature-Time Diagram. The power loss was calculated as:

$$P_{loss} = \frac{T_m - T_u}{R_{th}} = 7.52 \pm 0.82W \quad (14)$$

Therefore, the specific heat of fusion was calculated to be $11.65 \pm 2.3 \frac{kJ}{kg}$ with a 61% deviation from its literature value of $30 \frac{kJ}{kg}$.

The thermocouple might have been inadvertently set up in a way that caused it to measure the temperature of the crucible. That is possibly why the measurements differ from the expected values. The same might be the reason for the temperature for the melting phase not being constant. Though even if the sample had a constant temperature during this phase, the temperature of the crucible was constantly increasing.

Another reason could be impurities or non-uniform composition of the alloy. If there is an inhomogeneous distribution of elements in the alloy, it will no longer be a perfect eutectic mixture. This results in a non-uniform temperature in the melting phase.

6 Conclusion

This experiment demonstrated the relationship between heat transfer, phase transitions, and thermodynamic properties of metals. The results deviated strongly from the expected values, predicted by theory. This was most likely due to faults in the set up and measurement procedure.

Literature

Preparation Sheet, provided by Prof. Dr. Matthias Gramich, Prof. Dr. Daniel Förster, [Quantitative measurement of the specific heat capacity and heat of fusion of a eutectic alloy (SWE)]